The Intellectual Foundation of Information Organization

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above. However, they are not well designed for computer manipulation of this data. They have been criticized for their fixed-length, flat-file formats; for not presenting the kind of unified view of data required for efficient database design; and for being fraught with insertion, deletion, and update quirks that cause inefficiency in storing, maintaining, and performing retrieval operations on data.[37] To completely overhaul the MARC record, a standard that has been in force since the mid-1960s, is unthinkable for its economic, political, and bibliographical consequences. While MARC data can be refigured to streamline computer operations, consensus seems to be lacking on how to do this effectively and at a reasonable cost.

Given the burden put on it by the new media and the technological revolution, the traditional bibliographic record has so far remained surprisingly robust. This may yet change. Digital documents open the possibility of a radical deconstruction of the bibliographic record. Instead of gaining access to such documents indirectly through metadata on a surrogate record, access can be directly to the documents themselves, through the use of coding. Coding schemes, like the *Guidelines for Electronic Text Encoding and Interchange* and Standard Generalized Markup Language (SGML) guidelines, provide for identifying document attributes when and as they occur in the machine-readable text.[38] Where attributes essential for retrieval do not occur in the text, provision is made to supply them in document headers. The effect of this is to dispense with descriptions of documents as independent records and instead to distribute and embed the elements of description in the documents themselves.

An even more radical deconstruction is to dispense even with coding and to rely on documents to be self-describing. A system that relies on documents to be self-describing is founded on the supposition that the design of search algorithms can substitute for the crafting of individual bibliographic descriptions. The algorithms used in search engines today are still fairly primitive; many are based on keyword searching alone. As yet such systems have not been able to deal with the scatter and clog of information caused by the synonymy and homonymy of natural language, nor can they provide semantically useful displays of bibliographic data. But they have the potential to do so, transforming the theory of bibliographic description into a theory of bibliographic searching.

5

Principles of Description

Introduction

In the literature of bibliographic description the word *principle* has been used to refer to the objectives of a bibliographic system, to general rules in a bibliographic code, and to directives that guide the construction of a bibliographic language. It is the last sense that is used in this text, the sense Lubetzky used in noting that

> One cannot undertake to outline a code of rules without directive principles anymore that one could undertake to chart a map without a compass indicating the cardinal points. Nor can one approach and apply intelligently a code of rules, however well constructed, without a knowledge of its underlying principles . . . without a knowledge of its cardinal direction.[1]

This is also the sense of the word *principle* used in the International Federation of Library Associations and Institutions' (IFLA) *Principles Underlying Subject Heading Languages (SHLs),* where *principles* are defined as "general directives for determining the construction and application of subject heading languages."[2] More generally, viewed as a directive for action (designing a bibliographic language is an action), *principle* falls under the *Oxford English Dictionary*'s sense of "a general law or rule adopted or professed as a guide to action; a settled ground or basis of conduct or practice; a fundamental motive or reason of action, esp. one consciously recognized and followed."[3]

Bibliographic principles are different from bibliographic objectives and bibliographic rules. Objectives codify what a user can expect of a bibliographic system — to find a document, to find all manifestations of a work contiguously displayed, and so forth. Principles, on the other hand, are directives for the design of the bibliographic language used to create such

a system. This language normally takes the form of a code of rules. However, principles themselves are not rules but rather guidelines for the design of a set of rules.

Certain general principles govern the design of all systems, bibliographic and otherwise. Two of particular relevance to the design of a bibliographic language are the principle of sufficient reason and the principle of parsimony. Originally identified by Leibniz, and called by Ranganathan the canon of impartiality,[4] the principle of sufficient reason requires that the grounds for each design decision be defensible and not arbitrary. The principle of parsimony, another of Ranganathan's canons,[5] requires that where alternative ways exist to achieve a design goal, the way that best furthers overall economy is to be preferred.

In addition to general design principles are principles specific to the design of a bibliographic language. Those that have been recognized in the Anglo-American cataloging literature and followed in practice include the following:

- *Principle of user convenience* Decisions taken in the making of descriptions should be made with the user in mind. A subprinciple is the

Principle of common usage Normalized vocabulary used in descriptions should accord with that of the majority of users.

- *Principle of representation* Descriptions should be based on the way an information entity describes itself. A subprinciple is the

Principle of accuracy Descriptions should faithfully portray the entity described.

- *Principle of sufficiency and necessity* Descriptions should be sufficient to achieve stated objectives and should not include elements not required for this purpose. A subprinciple is the

Principle of significance Descriptions should include only those elements that are bibliographically significant.

- *Principle of standardization* Descriptions should be standardized, to the extent and level possible.
- *Principle of integration* Descriptions for all types of materials should be based on a common set of rules, to the extent possible.

User Convenience

The principle of user convenience requires that bibliographic descriptions be designed with the user in mind. It requires the user to be the focus in all

design decisions, no matter how minor.[6] Cutter, the user's greatest champion, elevated this principle to a categorical imperative in his often quoted dictum that "the convenience of the user must be put before the ease of the cataloger."[7] Cutter practiced what he preached. A striking example of this is his consideration for the unsophisticated user — children, the desultory user, and the man on the street who needs information quickly.[8] This consideration led him to reject the traditional European classified catalog, which was designed for scholars, and to create a whole new subject approach to information — an alphabetic approach that would allow unsophisticated users to get what they want by a name that they know, without any guesswork.[9]

It was in the context of an alphabetic-subject catalog that Cutter introduced the principle of common usage. He called usage "the supreme arbitrar,"[10] by which he meant that when things have alternative names (for instance, Cats and Felines), the choice between them should favor "the most usual name . . . , the one under which most people would be likely to look."[11] The common-usage principle was also recognized by Ranganathan, who called it the canon of the sought heading.[12] It is the cardinal principle governing the choice of terminology in all controlled vocabularies. In the current cataloging rules it is referenced by phrases like "commonly known," "predominantly identified," "most frequently identified," and "generally identified." The IFLA document on subject-heading principles calls it simply the "User Principle."[13]

Despite its preeminence as a directive, the common usage principle is both vague and unscientific.[14] Cutter himself recognized this. The problem comes with trying to ascertain what usage might be common. There is no single public, no user writ large; rather there are many users, and the way they verbalize their search requests represents many different language usages. In an operationalization of common usage, *AACR2R* interprets a "predominantly identified" form of name to mean "the form found in 80 per cent of the author's works."[15] This operationalization provides a decision rule; however, its validity can be questioned, as it represents the usage of publishers rather than that of the public. It identifies as "prominently identified" names such as Andersen, H. C. (rather than Andersen, Hans Christian) and Maugham, W. Somerset (William Somerset), rather than Maugham, Somerset.

The fact that language usage varies over place and time presents a dilemma insofar as vocabulary control has traditionally required a fixed usage. English-speaking users would expect to find Tchaikovsky's ballet listed as *Sleeping Beauty,* not under *Spîashchaîa Krasavitsa,* which is its internationally accepted uniform title. The fact that this is a dilemma is illustrated by a request once made to the British Library that it assign two uniform titles to works: one formulated to conform to the international standard and the other formulated to be familiar to local users.[16] Relevant to this dilemma is another of Ranganathan's canons, that of local variation, which requires that the needs of local constituencies not be ignored.[17] In a digital environment, it is relatively easy to honor this canon as it applies to usage, since such an environment permits the automatic linking of various forms of names. Automatic or transparent linking allows unity to be achieved without sacrificing diversity. If all the different names by which something can be called are linked transparently, so that whichever a user chooses will make a direct match in retrieval, all usages are served, common and otherwise. Technological advance makes it possible to replace authority control with what Tillett calls "access control."[18]

During the last third of the twentieth century the principle of user convenience and its subprinciple of common usage have been the objects of research. Dozens of studies have been undertaken to determine how users search for information and what problems they encounter. While the existence of such studies underscores the importance attached to the principle of user convenience, the findings themselves are not unexpected. One finding is that users tend to use a very limited set of metadata, usually those found in bibliographic citations, such as author, title, edition number, publisher, place of publication, and date.[19] This finding could have been predicted as an instantiation of the general law that human selection from any kind of store of items tends to follow a Bradfordian distribution.[20] Another finding is that in subject searches users tend to retrieve too many or not enough citations,[21] which, again, is not unexpected insofar as it is anticipated by, and its antidote hypostasized in, the collocation objective. Another finding is that users have trouble matching their own vocabulary against that of the retrieval system they are using.[22] This is not surprising as it illustrates the above-mentioned, well-understood problem of attempting to determine a single common usage.

Representation

The principle of representation requires bibliographic descriptions to be constructed to reflect the way bibliographic entities represent themselves. This principle is used primarily to ensure accuracy of description, though it is used as well to contain costs, to prevent idiosyncratic descriptions, and to assist in the construction of operational definitions.

Accuracy

A description is inaccurate if it in any way misrepresents an entity, making it seem what it is not. In bibliographic contexts, accuracy, in the sense of honest portrayal, is equated with truth in transcription.[23] Truthfully transcribing how a document represents itself is necessary for the identification and communication of bibliographic information. The metadata in a description are truthfully transcribed, or warranted, if (1) they come from a specified source and (2) they are copied in the form in which they appear there, except for capitalization and punctuation. These two conditions ensure the likelihood that two people describing the same bibliographic entity will create identical descriptions.

Prescribing a source of information from which data elements should be derived is a way of specifying how an entity can represent itself. In simpler times, when bibliographic entities were for the most part books published in Western countries, the choice of source was obviously the title page, the "face of the book."[24] In 1884 Charles Blackburn wrote: "It is I believe one of the laws of bibliography that catalog titles should be copies of the title pages of books." [25] Lubetzky wrote regarding the title page that "the most important characteristic of a book for the purposes of cataloging is the fact that it is provided with a prominent identification tag in the form of a title page."[26] Ravilious expressed the same thought in referring to a title page as "a magnet at which bibliographic minutiae collect like iron filings."[27] Cutter spoke of "the cult of the title page" and how "its slightest peculiarities are noted; it is followed religiously, with dots for omissions, brackets for insertions and uprights to mark the ends of lines; it is even imitated by the facsimile type or photographic copying."[28]

The title page of a book is a reliable source, since it, together with its verso, usually contains all bibliographically significant data. Most nonbook

materials, however, lack a convenient identification tag in the form of a title page. This difficulty has been met by specifying for each of the different kinds of nonbook materials title-page analogs — that is, the chief source from which data can be taken. Though the sources vary, depending on material type, an attempt at uniformity was made by specifying common criteria for the selection of sources. In 1975 C. P. Ravilious identified three such criteria as a result of surveying codes specifically geared to the description of nonbook materials.[29] These criteria, along with some of the problems attending them, are

- *Comprehensiveness* Given a choice of several sources, data are to be taken from the source that gives the fullest, clearest, and most authoritative information — that is, from the source that most adequately identifies the item. A problem with this criterion is that sources do not always behave in such a manner that it is possible to identify one that is most comprehensive.
- *Proximity* Given a choice of several sources, data are to be taken from the source that is nearest the item's content, such as an internal source. There are at least three reasons why this criterion is an uncertain guide: (1) the data on the item itself may not be complete, (2) the data may not be authoritative, and (3) it may be difficult or time consuming to get to the data (for example, when the source is unreadable without processing, as is the case with compressed or printer-formatted data in an electronic resource).
- *Persistence* Given a choice of several sources, data are to be taken from the source that is most enduring (for example, from the item itself) rather than its packaging. Taken literally, this criterion would require taking data from a sound recording rather than the label affixed to it and from the title and end frames of a video or from the internal screens of an electronic resource, rather than from containers for these items, labels on the items, or accompanying documentation. This makes the criterion subject to the same problems as those afflicting the proximity criterion.

With modification, Ravilious's criteria were used for the various nonbook International Standard Bibliographic Descriptions (ISBDs) to develop a preference ranking of sources for bibliographic data. But ranking does not address the major problem, which is that often no single source is comprehensive. Data from a chosen source can always be supplemented with data from other sources, bracketed to signify their nonstandard origin. There is a danger, however, of too often gravitating toward the default position, where data are taken from anywhere in an item — in effect, treating several sources as a unitary source. This could imperil accuracy (the reason

for prescribing a source in the first place), since a given datum, such as a title, can appear in different forms in different places on a document. The only way accuracy could be ensured would be to accompany each datum with a location-where-found annotation. Truth in transcription was once relatively easily ensured by faithfully copied title pages, but with the influx of nonbook materials it has become troublesome and has raised to a theoretical level questions about the limits and purpose of accuracy and the conditions required to ensure it.

In some cases data may be accurately transcribed from a document, and yet the resulting description is not truthful. For instance, a document may announce itself as a second edition, whereas in reality it is only a reprinting of a first edition; or it may purport to be by a given author, whereas in reality it is the work of a ghost writer; or it may be attributed to a corporate body and yet be written by a person. What is to be done is such cases? Lubetzky suggests that one can be guided by the principle of representation:

> It must be recognized that it is really not "the writer of a book" or the creator of a work who will generally be regarded as the author or the one "chiefly responsible" for it, but the one who lent his name and authority to it — the one *represented* as the writer of the book or as the creator of the work, who presumably formally assumed responsibility for it.[30]

When bibliographic reality conflicts with existential reality, the former has the stronger claim. The reason for this is primarily economic. In-depth bibliographic research of the sort that pursues questions of attribution is a luxury affordable only in special situations, such as in the description of art works or rare books, not in the day-to-day work of introducing order into the bibliographic universe. Some small indulgences are permitted. When ambiguous, unintelligible, or obviously inaccurate statements appear on documents, these may be explained or corrected, within brackets and to a reasonable extent.[31] Nevertheless limitations must be set, and here the principle of representation becomes an injunction to eschew brooding over the true nature of a bibliographic entity — and, not incidentally, to eschew the temptation to make idiosyncratic descriptions.

Definitional Role of the Principle of Representation

In chapter 2 it was seen that difficulties in defining terms such as *edition* and *author* can sometimes be resolved by enumerating character strings on documents that are indicative of editionhood or authorship. A given

document is a second edition if it says it is. A person is an author if that person is represented as such. The principle of representation thus has a role to play in resolving definitional ambiguities. In those cases where it is not clear how an entity *(x)* represents itself, conditions can be enumerated under which *x* is represented as *y*. For instance, in the classic case of a report prepared by a personal author and issued in the name of a corporate body, conditions are stated under which the report can be said to emanate from the body — for example, the report is of an administrative nature or records the collective thought of the body. A fully operational definition would go a step further to specify just which character strings on the report are indicative of "being of an administrative nature" or "representing collective thought." Presently there is considerable interest in how bibliographic information is represented on documents, and this may be expected to continue. Both standardization and automation assume that a document is what it represents itself to be and that clues in and on them can be used to infer in a mechanical or semimechanical way what a document's title is, who its authors and publishers are, and even what it is about.

Limitations of the Principle of Representation

The principle of representation if carried too far would result in bibliographic descriptions that individually are correct but when taken collectively are inconsistent. It would go against the principle of user convenience if bibliographic records — say, for books — were literal copies of their title pages. Title pages vary considerably. They contain differing amounts of information, not all of it of interest; the order in which the information is presented varies, with the author appearing either before or after the title; different conventions are used for punctuation, abbreviation, and capitalization; and so on. If bibliographic descriptions mirrored title pages, the information they impart would be difficult to grasp quickly; the eye would never know where to find a given kind of datum, such as an author or title. To facilitate scanning, certain aspects of descriptions need to be made uniform. These include the selection of data elements, the order in which they are displayed, and their punctuation, capitalization, and abbreviation. Thus, the principle of representation is curtailed by the need for consistently formatted bibliographic records.

Accuracy in description does not by itself ensure effective retrieval. Thus, the principle of representation is of limited scope in that it deals only with those aspects of description based on derived vocabulary. Descriptions cannot consist of derived data alone because the language in which a document represents itself is not necessarily the one a user would use to look for the document. A user cannot be expected to guess the name or form of name by which an author is represented on a document. To gear the language of users to the language of documents, a normalized vocabulary must be introduced to supplement vocabulary derived from documents. The principle of representation is useful in developing means to achieve the finding and identification objectives, but here its usefulness stops.

Sufficiency and Necessity

Since Aristotle's time, philosophers have debated the nature of description, attempting to distinguish the attributes of things that are essential from those that are accidental. In the context of bibliographic description, the endeavor is the same; the attempt here is to distinguish attributes that are bibliographically significant from those that are not. The directive that guides determinations of bibliographic significance is the principle of sufficiency and necessity. This principle requires a bibliographic description to be full enough to meet the objectives of a system for organizing information and, at the same time, contain no data elements extraneous to these objectives.

Panizzi implicitly tied considerations of sufficiency and necessity to the bibliographic objectives when he argued that descriptions must be full enough to distinguish one edition of a work from another and to enable a choice to be made from among similar editions.[32] Lubetzky made the connection explicit. Up until the 1940s it was generally assumed that the description of a book should faithfully reproduce all data elements on its title page. Lubetzky objected to this, arguing that objectives, not title pages, should determine what was and what was not essential in a description. If data elements on a title page did not further a bibliographic objective, they were to be excluded; if data elements were not on a title page and yet were necessary to meet one of the objectives, they were to be introduced.[33] Relying on objectives rather than on an unthinking adherence to the principle of representation was a revolutionary concept at the time (1946) and

was opposed on the ground that omitting title-page elements in a bibliographic description would make it impossible to distinguish various editions of a work. This argument proved to be not valid.[34]

Nevertheless, a problem arises in referencing bibliographic objectives to determine essential data elements, at least as the objectives are currently formulated. The problem is caused by the open-ended objectives. The choice objective in encouraging the inclusion of any data element that might be useful in the selection of a document can lead to the proliferation of expensive detail. The same is true of the navigation objective and even the collocation objective. As was earlier observed, containing bibliographic detail needs to be addressed at the level of objectives — in particular, the open-ended objectives. Only when this is done can the objectives be used to settle questions of necessity and sufficiency.

Cutter was one of the first to face the problem of necessary versus unnecessary data elements. He viewed appropriate fullness of description as a variable, depending on local needs and circumstances, such as library size. Accordingly, he distinguished three kinds of catalogs — short, medium, and full. This distinction continues to be carried in cataloging codes to the present day, but it is becoming less useful as global cataloging renders distinctions based on library size irrelevant. The goal of reducing bibliographic effort so that one item need be described only once worldwide necessarily assumes adherence to a uniform level of description, one deemed adequate for national libraries.

Occam's Razor

The principle of sufficiency and necessity at times acts like the principle of parsimony, a brake limiting the metadata admissible in description. It is generally, though not quite accurately, assumed that the cost of making descriptions increases in proportion to the number of data elements they contain.[35] This leads to the conclusion that data should not be multiplied unnecessarily, which in turn leads to taking up Occam's razor and slashing elements deemed bibliographically insignificant. Decisions as to which these are have been justified on various grounds, among them reason, empirical warrant, and expert opinion.

The most celebrated wielding of Occam's razor followed on Lubetzky's "Is this rule necessary?" campaign of 1953.[36] Having achieved earlier success in eliminating title-page data not required by the bibliographic objectives, Lubetzky, as a further step in his reform to simplify cataloging, set about eliminating redundancy in the rules for providing identifiers for bibliographic entities. Observing that many of these rules were designed to fit specific cases rather than general conditions,[37] he subjected them one by one to stringent review. Was a given rule part of a larger rule already included in the code? If so, it was redundant and had to be eliminated. Was it consistent with other rules? If not, it had to be revised. A tour de force of rational argument, Lubetzky's campaign was successful, and the consequent simplification of the rules for identifying bibliographic entities was heralded as a return to principle.

The problem how to simplify bibliographic description by eliminating unnecessary metadata has been approached through empirical research. Following Lubetzky's call to eliminate unnecessary title-page data elements, a study was conducted at the Library of Congress to assess the effectiveness of page count in distinguishing one edition of a book from another.[38] (The study was needed to refute the argument that full title-page transcription was necessary to identify a document.) Though the sample was small, page count, along with author and title data, did in fact prove a reliable indicator of bibliographic uniqueness. Half a century later, in a bibliographic world significantly more complex, Ed O'Neill and others conducted a similar study at the Online Computer Library Center (OCLC).[39] For the purpose of weeding out duplicate records on the OCLC database, they developed a duplicate detection algorithm based on thirteen weighted data elements, which they then tested for efficacy. These data elements proved to be sufficient for meeting the identifying objective for a bibliographic database of over 36 million records.

Empirical research has also been used to explore the necessity and sufficiency of bibliographic data elements from a users' point of view. The Seal study mentioned earlier, which surveyed catalog use in a polytechnic library of 320,000 books and 160 nonbook materials, pitted an experimental catalog containing only brief bibliographic records against a normal catalog containing full records.[40] Participants in the experiment were asked first to search in the experimental catalog and then, only if they could not find what they wanted, to use the normal catalog. It was found that in 92 percent of the searches there was no need to resort to full records.

Several observations can be made about this and other such studies. First, as already has been observed, the results are to be expected in that the frequent use of a few data elements obeys the Bradfordian law of selection. Second, because of their limited scope and the simplifying assumption that only user-identified data elements need be included in descriptions, the studies lack external validity. In a global bibliographic environment, the act of uniquely distinguishing one information entity from another is no longer a local matter. The few metadata elements needed to distinguish 480,000 documents hardly suffice to distinguish a hundred times that many. Third, such studies, in their intent to understand the user, as represented by J. Q. Public, overlook the often more complicated needs of another class of users — those involved in database construction and maintenance. J. Q. Public may seldom need to distinguish nearly identical documents, but acquisition librarians need to do so on a daily basis. Moreover, even a sophisticated J. Q. Public may have only a limited understanding of his needs, unaware of the guidance that is given by full rather than abbreviated bibliographic records.

The most usual approach to data element elimination has been to rely on expert opinion. An example is the expert opinion lodged at the Library of Congress and promulgated in its rule interpretations. Many of the Library of Congress rule interpretations (LCRIs) are indeed interpretations in the sense of clarifying ambiguity, but some have the sole purpose of cost-cutting. LCRI 1.1F1 limits statements of responsibility by ruling on the bibliographic significance of persons whose names appear on a document (book-jacket designers are out). LCRI 2.5C1 and 2.5C2 override the *AACR2R* rule suggesting that various types of illustration be named (such as coats of arms, facsimiles, forms, genealogical tables, maps, music, plans, and portraits), recommending instead that all illustrative matter regardless of type be indicated by *ill.*[41]

Because of the need to standardize bibliographic descriptions, the expert opinion relied on for decisions of sufficiency and necessity frequently is lodged in political bodies like task forces working under the aegis of national and international associations. Through cooperative efforts such bodies work toward a consensus on minimal data-element sets. An example is the Program for Cooperative Cataloging's core-level record standard for books.[42] Others are the Cooperative Online Serials Program (CONSER) standard for serials[43] and the data-element set produced through an exercise in bibliographic modeling by the IFLA Study Group on Functional Requirements for Bibliographic Records.[44]

The principle of sufficiency and necessity can be used to justify either expanding the number of data elements in a description, as was done by Panizzi, or reducing them, as was done in the polytechnic experiment. Decisions relating to bibliographic significance are difficult but are not entirely a matter of the opinion of users or experts. An objective determination can be made of the data elements needed to identify a bibliographic entity and distinguish it from other like entities.[45] It is where the objectives are open-ended that decisions must be left to those who are able to balance user needs and bottom lines.[46]

Standardization

It has been observed that "the entire history of bibliographic control is that of the establishment of standards and their subsequent modification."[47] In the mid-nineteenth century Charles C. Jewett saw standardization as a necessary condition for the union catalog he was advocating to be housed at the Smithsonian Institution. He envisioned this catalog as containing records for the holdings of all public libraries in the United States. Such a catalog would be an instrument for national bibliographical control, facilitate the exchange of bibliographic information, and reduce duplication of effort through cooperation. For such a catalog to be made actual, bibliographic records would have to be constructed according to a uniform style. Accordingly, he wrote: "The rules for cataloguing must be stringent, and should meet, as far as possible all difficulties of detail. Nothing, so far as can be avoided should be left to the individual taste or judgment of the cataloger."[48]

Jewett did not live to see a union catalog realized. The requisite technology to mass produce and distribute bibliographic records did not exist until card catalogs began replacing book catalogs. The year the Library of Congress began its card-distribution program, 1901, marks a technological advance and the beginning of union catalog building in the United States. During periods of technological advance, cooperative fervor and activity tend to accelerate, and the drive toward standardization is strong.

The beginning and end of the twentieth century were two such periods — first when card catalogs were introduced to replace book catalogs, and then again when online catalogs superseded card catalogs.

To standardize means to bring into conformity with "something established by authority, custom or general consent."[49] A natural language is only partially standardized through the force of custom; a bibliographic language, on the other hand, is strictly standardized through the general consent of those involved in building international codes. The development of a standardized lingua franca for bibliographic description is a twentieth-century achievement, due in large part to IFLA's ability to provide the infrastructure by which representatives from countries worldwide could meet and hammer out consensual accords.[50] A milestone in IFLA's standardization activities was the 1961 International Conference on Cataloging Principles. At this conference fifty-three countries and twelve international organizations met to agree on principles for the selection and normalization of metadata for accessing bibliographic information. Eight years later, another important conference, the International Meeting of Cataloging Experts (IMCE), led to the establishment of international standards for bibliographic description. Eleven years after this still another historic international meeting was convened, this time to develop a universal Machine Readable Cataloging (MARC) format (UNIMARC).

Standardization admits of degree. Some, such as Jewett and the IMCE experts, would argue that the international exchange of bibliographic information requires maximum standardization of bibliographic descriptions,[51] but others are concerned that standardization can be carried to excess. One danger of excessive standardization is conflict with the principle of user convenience. Cutter wrote that

> strict consistency in a rule and uniformity in its application sometimes lead to practices which clash with the public's habitual way of looking at things. When these habits are general and deeply rooted, it is unwise for the cataloger to ignore them, even if they demand a sacrifice of system and simplicity.[52]

Prophetically he observed that "no code of cataloging can be adopted in all points by everyone,"[53] an observation echoed fifty years later by J. C. M. Hanson: "Should an international code ever become a reality, many librarians would refuse to subscribe to it in its entirety."[54]

A second danger of standardization carried too far is that the reasons and principles underlying a bibliographic code become obscured. Andrew

D. Osborn argued this point, using the term *legalistic* to damn a code that attempts to include rules and definitions to govern every point that arises.[55] Lubetzky did as well, finding himself unable to "view with equanimity the continuous proliferation of the rules, their growing complexity, and the obscurement of the objectives and design of the code as a whole." This led him to ask his riveting question: "Are all these rules necessary? are all the complexities inevitable? is there an underlying design which gives our code unity and purpose?"[56] To which his answer was that cataloging should be done by principle rather than by a slavish following of rules. Paul Dunkin in his *Cataloging U.S.A.*, a history of Anglo-American cataloging, relates how over the years the pendulum has swung between the opposing forces of legalism (the law, as represented by Jewett) and romanticism (the prophets, as represented by Cutter, Osborn, and Lubetzky).[57]

A third danger of overstandardization is its tendency to inhibit change. Bibliographic change is necessarily conservative. Where technological, political, and economic forces make change seem desirable, expense of money, spirit, and effort is often sufficient to counter it. For example, while it might be desirable to modify the present MARC format to make it more suitable for communicating bibliographic information and for database design, so entrenched is this standard, so hard-won through years of cooperative effort, that to set in motion the elaborate and time-consuming political mechanisms needed to effect a major change is hardly an option. Where changes to bibliographic codes and formats are introduced, they are normally minimal and incremental, which sometimes leads to sacrificing conceptual consistency for a quick Band-Aid fix.[58]

Given that standardization in bibliographic description can be excessive, the question arises of how to limit it — how to determine just which aspects of description need to be standardized. Ron Hagler suggests limiting standardization to only those aspects that are essential for the identification of works and documents. While the rules governing these should be prescriptive, those governing other aspects of description could well be discretionary.[59] The suggestion has merit, not the least because it can be acted on, since it is in fact possible to ascertain those data elements essential for identification.[60]

One aspect of bibliographic description that does not require standardization in the traditional sense has already been mentioned. This is the standardization achieved by normalizing of names for works, authors, and

subjects. Whereas in a card-catalog environment, normalization required the choice of one authoritative name, in an digital environment, transparent linking allows authorized names to be replaced by compatible names.[61] This does not mean that rules for normalizing can be dispensed with entirely but only that different rules are needed to ensure a uniform practice. In any case, the principles of standardization and common usage need no longer conflict. Insofar as universalism in any endeavor is furthered by the preservation of diversity, standardization that does not abrogate the need for local variation is a giant step forward in universal bibliographic control.

Integration

The principle of integration is a directive to use a common set of rules for all media, with exceptions being made only for material-specific attributes. Both the principle of standardization and the principle of integration mandate uniformity in description — the former across space and time and the latter across material type. Uniformity in describing bibliographic entities, irrespective of the medium in which they are embodied, is desirable for several reasons: it serves the objectives of the catalog, particularly the collocating objective; it serves user convenience in providing a common interface to bibliographic information; and it serves the principle of parsimony in achieving economy of expression.

Since Panizzi's time, descriptive codes have had to deal with documents in some way out of the ordinary — for example, those using typography other than alphanumeric characters, such as musical notation, or those not packaged in book form, like a single map. For documents such as these, special rules had to be developed. Panizzi and Cutter both sought the help of experts to develop rules for music materials. In the 1940s and 1950s, the Library of Congress also turned to specialists to draft rules for its growing collections of motion pictures, sound recordings, and pictures. The Library of Congress rules proved difficult to use and, as a result, were rejected by most school and public libraries. This led to a proliferation of locally developed manuals to describe nonbook materials, simultaneously abrogating the standardization principle and that of integration.

Early in the 1970s reaction set in, and a swing began away from specialization and toward integration. Committees first in Canada and then in England and the United States began to formulate rules for nonbook materials that would be compatible with those used for books. As might be expected, some opposition to this was voiced, with one critic complaining that "what was good for monographs and serials suddenly seemed equally desirable for cartographic materials, old or rare books, music and nonbook materials in general."[62] Particularly strong opposition came from the Association for Educational and Communications Technology, which, though it has since come round somewhat, still today favors an anti-integration stance.[63] Despite the opposition, the drive toward integration gathered steam, culminating in the early 1980s with the publication of *AACR2,* which was hailed as "a major breakthrough" in integration.[64] Belief that integration was a good thing elevated it to a principle.

The attempt to extend a bibliographic language originally designed to describe documents in one medium to encompass all media tests the theoretical soundness of the language. The language becomes strained as problems are encountered and its flexibility is challenged. Some problems are relatively easy to solve. The fact that nonbook entities lack authoritative sources of information such as title pages is a problem; but (as has been seen) means for designating title-page substitutes can be devised. The fact that many nonbook entities lack titles is a problem;[65] but this can be dealt with by generalizing already existing rules devised for books that lack titles. The fact that many nonbook entities require special equipment to be experienced is a problem; but it can be handled by requiring technical specifications and summaries in bibliographic descriptions.

Problems not easy to handle in generalizing a book-oriented language are definitional in nature. The definitions of *publication* and *publisher* are examples of this. The concept of *publication* may be extended beyond its traditional connotation of formal publication to its broader etymological meaning of "to make public" and that of *publisher* to "any release agent." However, doing so results in the loss of useful distinctions, spawns difficult-to-decide cases, and causes inconsistencies. Take the case of electronic documents. The *International Standard Bibliographic Description for Electronic Resources (ISBD(ER))* considers these to be published.[66] But it is a question whether they are any more "made public" than manuscripts and original works of art, which, though they may be viewed by many people, are not technically published.

As for the concept of publisher, when defined broadly it assumes a chameleon quality. For instance, if an electronic document is not identified as having a formal publisher (in a traditional sense), the author, regarded as "release agent," might be taken as publisher; if the same document happens to bear not only the author's name but also that of a distributor, then the distributor might be taken as publisher; if the same document bears the name of the author, distributor, and also that of a publisher — well, of course, the publisher is the publisher. But when the concept of publisher varies with what happens to be written on the document, it begins then to lack intrinsic meaning.[67]

Even more serious are the definitional problems attaching to the concepts of work and edition. The difficulty of interpreting these concepts for nonbook materials (discussed in chapter 3) is the kind that strikes at ontological commitment and shakes theoretical foundations. If it is not clear how these concepts are to be defined, then it follows that it is not clear what is meant by the bibliographic objective that calls for bringing together the various editions of a work. In other words, it is not clear what it is that is supposed to be organized.

The concept of author is also problematic in translating a book language to a multimedia language. For the century and a half since Panizzi, it has been assumed that users search for information by author. However, this assumption is not valid for many nonbook materials, particularly those of mixed authorship that involve numerous people performing different kinds of functions. In the creation of moving-image materials, for instance, many people are engaged in a variety of artistic and intellectual roles, no one of whom is an author in the commonly understood sense. For these and other nonbook materials, the concept of author wobbles, and when this happens, so too does the objective that requires collocation by author. When in a theory something as high-level as an objective begins to lose some of its universality, then the foundation of that theory begins to crack.

It is possible that the definitional problems raised by nonbook materials are serious enough to tilt the integration-nonintegration balance back toward nonintegration. An indication that this may be happening is another acceleration in the publication of manuals for nonbook materials. While these purport to be interpretations of the *Anglo-American Catalog Rules,* second edition, revised *(AACR2R),* they clearly extend its

reach. It remains to be seen whether this phenomenon is significant enough to represent a move away from a common bibliographic language that integrates rules for documents in all media, to a family of loosely related, medium-specific languages.

The principle of integration, like the principles of user convenience, representation, sufficiency and necessity, and standardization, functions as a directive guiding the construction of bibliographic languages. Principles direct in the sense of providing a rationale for design decisions. Other things being equal, where there are design alternatives, one that references a principle is to be preferred to one that does not. But this is obvious — a textbook maxim. Frequently other things, notably costs, are not equal, and frequently one principle vies with another in decision making. The following chapters on the design of specific languages illustrate conflicts of this sort and provide additional examples of how principles function — their usefulness, viability, and internal conflict — in bibliographic description.